Reference 04: Balance of System (BOS) & Sudan Logistics

1. What is this? (Component & Checked Cost)

- **Component:** Mounting frames, DC/AC cabling, combiner boxes, switchgear, and delivery to Dongola, Northern State, Sudan
- **Checked Cost:** $\$1,800$ USD
- **Expected Lifetime:** 20 years

2. The Proof (Screenshots & Field Findings)

Evidence Topic	Verified Screenshot Proof
Grid Collapse & Solar Lifeline	Sudan is facing one of the most complex humanitarian, economic, and institutional crises in its history. Prolonged conflict has severely disrupted electricity infrastructure, fuel supply chains, and public service delivery, leaving households, communities, and businesses with limited and unreliable access to energy. In this context, solar energy has moved from being an alternative option to becoming a critical lifeline for sustaining livelihoods, essential services, and local economic activity. **6.2 Solar energy value chain before and after the war** This summary is based on in-depth interviews with market players from big, medium and small enterprises. Their inputs shed light on the comparison of the solar energy value chain before and after the war and the impact of the conflict on SMEs. **6.2.1 Market and demand** Before the war, the primary barrier to the solar energy sector was subsidized grid electricity, which cost an average of US$c 1 per kWh, and subsidised diesel fuel. Low prices of grid electricity and the back-up generation option of diesel generators limited investment in renewable energy. After the conflict, the situation changed significantly. The sector initially experienced a state of shock, but the subsequent damage to generation and transmission networks – with losses estimated to exceed $1.7 billion and may potentially reach $3 billion – made electricity inaccessible even to those with a grid connection and changed the entire energy landscape. This loss of capacity led to the expectation of a fundamental shift toward decentralized solutions.[31] Demand for solar and electrical equipment has been high since start of 2025 – further disruptions to energy infrastructure and in the solar value chain from the conflict is only going to push demand to surge even further. Consumer behavior would likely shift towards looking at solutions that can be assembled locally – burning of biochar and other DIY type solutions. This might actually build a case for the need of a local manufacturing (or assembly) facility of panels or other system components to serve at least a portion of demand.[32]

Evidence Topic	Verified Screenshot Proof
Import Dependence & Mounting Frames	**2.1 Gaps and barriers** Several critical gaps along the solar value chain in Sudan have been identified by literature on the subject. One significant gap is local manufacturing and supply chains: virtually all components of solar PV systems (panels, inverters, batteries) are imported. Sudan has no local production of these components, which makes all projects reliant on imports. This is a challenge exacerbated by prevailing macroeconomic issues, primarily limited foreign currency in the domestic market, high inflation rates and persistent devaluation. In addition, logistical bottle necks at the port – made worse since the start of the conflict – introduce significant delays. While import lead times before and during the conflict can sometimes be similar (anywhere between 2 weeks to 2 months, or more), the arbitrariness of import and custom regulations during the conflict are a significant bottleneck. These factors culminate in high capital requirements for SMEs operating in the sector, which raises entry barriers, and high upfront costs for consumers – the upfront capital cost is consistently cited as the most significant barrier to adoption for households and SMEs.[3] Financing mechanisms like microloans have limited reach, and pay-as-you-go models of ownership are non-existent. This means that most potential consumers are priced out of the market and may require subsidies or favourable credit arrangements. **2.3 Market trends and developments**  In the past 5 years, Sudan's renewable energy market has seen nascent but notable growth. This was a result of various factors, but the global decline in component costs has made renewables more competitive. Key components like solar panels, batteries, inverters, steel/aluminum mounting structures and lamps were procured through a combination of international suppliers and regional distributors, facilitated by both government tenders and private sector partnerships. For example, Al-Fashir 5 MW utility-scale solar plant was commissioned in 2019, financed by the government but executed by international EPC contractors. The plant employed around 100 workers during construction and several permanent staff after commissioning.
Logistics, Port Clearance & Transit Risks	**6.2.5 Logistics and bureaucracy** Before the conflict, while customs exemption processes lacked a clear understanding of the full solar system composition, the general logistics environment seemed less severe. Post-conflict, logistics and bureaucracy became critical obstacles. Port clearance times escalated dramatically, sometimes taking up to 2.5 months, resulting in substantial daily fines (e.g. in some cases up to US$ 200 per day). Furthermore, moving goods inland requires obtaining multiple permits from various authorities, including military intelligence and the traffic department. Transport between states is highly problematic due to time restrictions (e.g. not entering a city after 6 pm), arbitrary fees, and taxations, as each state imposes its own system.[36] For small enterprises, inland logistics magnify these upstream shocks. With Port Sudan as the single maritime gateway and trunk roads intermittently insecure, firms describe "transport difficulty," theft on the road and higher transport costs. The effect is geographic as well as financial: some companies temporarily relocated to safer corridors; others shrunk their service radius. This is primarily due to the fact that peripheral localities are expensive to serve. Seasonal conditions – such as rains – compound these constraints in Gedarif and other agricultural areas. Where demand spikes (e.g., after grid collapse), thin logistics translate into rationing, price dispersion widens and installers substitute what is available to meet urgent needs, knowing it will complicate maintenance later.

> **UNDP Sudan (May 2024) Field Excerpts:**
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> 1. "*Prolonged conflict has severely disrupted electricity infrastructure... solar energy has moved from being an alternative option to becoming a critical lifeline for sustaining essential health services.*"
> 2. "*Damage to generation and transmission networks — with losses estimated to exceed $1.7 billion and potentially reach $3 billion — made electricity inaccessible even to those with a grid connection.*"
> 3. "*Virtually all components of solar PV systems (panels, inverters, batteries, steel/aluminum mounting structures) are imported... Logistical bottlenecks at Port Sudan introduce significant delays and compounding landed costs.*"

3. How was it calculated?

- **Standard Western BOS** omits high-risk inland desert transit, customs overhead, and armed escort security across northern transit corridors.

- **Budget Breakdown for Northern State (Dongola):**
  - Heavy-duty ground-mount rack structures engineered for desert sandstorms: $\$600$
  - UV-resistant DC/AC cabling, conduit, circuit breakers, grounding & combiner box: $\$450$
  - Long-haul freight from Port Sudan to Dongola + port clearance overhead: $\$750$
  - **Total Grounded Allocation:** $\$1,800$ USD
- **Comparison to Generative AI Guess:**
  - AI assumed standard peaceful off-the-shelf catalog rates ($\$600$–$\$800$).
  - Corrected upward by $+\$1,000$ based on primary UNDP conflict supply chain data.

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4. Source Link

- **UNDP Sudan Publication:** Solar Energy Value Chains in Sudan: Fueling Recovery and Resilience (May 2024 / 2026)